

Stock Market Predictions

Risk Management

Sales Forecasting

Trading

AI Finance

Lecturer: 정재식 (GN 805, cschung7@gmail.com)

Class Time: Wed. 13:30–16:00

조교 (Teaching Assistant): 원유석 (wys0225@sogang.ac.kr)

Credit & Loan Decisions

Anti-Fraud Systems

Personalized Banking

Process Automation

Course Overview

Course Objective This course integrates finance with AI coding. Students are required to use AI-assisted coding tools from the categories below.

IDE

AI-assisted coding environments

cursor

codex

claude-code

ADE

Agentic dev. environments

Hermes

Grok-Bot

Buzz

Prerequisites Basic Python tools:

numpy

pandas

matplotlib

NO AUDITING ALLOWED

Covered Topics — Finance

■ Information Bars

- Price bars, volume bars, dollar bars, and tick bars

■ CAPM, Three-Factor Models, C-CAPM

- Asset pricing models and consumption-based approaches

■ Mean-Variance Frontier and Asset Allocation

- Portfolio optimization and efficient allocation strategies

■ Market Microstructure

- Bid-ask spread, trading volume, and private information

Covered Topics — Machine Learning

Selected Algorithms

Key machine learning methods for financial applications

Naive Bayes

Linear Regression

Lasso Regression

SVM

Clustering

Gaussian Mixture

Random Forest

Large Language Models

Applications of LLMs in financial markets and analysis

Network Analysis

Introduction to network analysis in financial contexts

Course Management

Programming Language: Python

Course Structure

Weeks 1–6

- 2.5 hours of lecture
- 0.5 hours of Q&A and hands-on coding practice

Weeks 7–10

- 2.5 hours of lecture
- Presentations on selected topics (meta-strategy, portfolio, LLMs)

Weeks 11–12

- 1.5 hours of lecture
- Student presentations on empirical financial research

Requirements

- Each student must write a **term paper** related to economics or finance using **big data** from open-source platforms.
- **Python** is strongly recommended. However, students may use other languages upon consultation.
- Projects involving **LLMs**, such as domain-augmented knowledge retrieval, are highly encouraged.

Textbook and References

■ Main Textbook

- Marcos López de Prado, *Advances in Financial Machine Learning*, Wiley (2018)

■ Lecture Materials

- Lecture notes in .py, .md, and .pdf format will be distributed before class

■ Programming Resources

- Python libraries: `numpy`, `pandas`, `matplotlib`, and AI-assisted coding tools

Grading Breakdown

Homework / Midterm 30%

Presentation 50%

Term Paper 20%

Term Paper

Each student must write a term paper related to economics or finance using big data from open-source platforms.

Projects involving LLMs, such as domain-augmented knowledge retrieval, are highly encouraged.

Term Paper Timeline

September 23, 2026 (Wed)

Term paper outline due — research topic, methodology, expected outcomes

October 7, 2026 (Wed)

Term paper 초안 발표 (draft presentation) — present the draft in class for feedback

October 21, 2026 (Wed)

Basic statistics and preliminary results — initial data analysis and findings

November 11, 2026 (Wed)

First draft due — all sections, analysis, and conclusions

Course Schedule

Week(s)	Topic
1–1.5	AI Finance Overview & Environment Setup
2	AFML Ch.2: The Information Bar (price, volume, dollar bars)
3	AFML Ch.3: Triple Barrier and Meta-Labeling
4	AFML Ch.4: Financial Data Challenges
5	AFML Ch.5: Long Memory and Feature Engineering
6	AFML Ch.6: Ensemble Methods and Applications
7	AFML Ch.10: Automatic Bet Sizing
8	AFML Ch.19: Private Information & Feature Engineering
9	NLP Feature Engineering and LLMs
10	Fine-tuning, Contextual Learning, and Applications
12–13	AI Agents and Their Applications in Financial Markets

AFML = Advances in Financial Machine Learning (Marcos López de Prado)